for nowadays. Adenosine triphosphate and the role of mitochondria and other cell structures, as well as the function of deoxyribonucleic acid in relation to genetics, are mentioned, for example. However, on the whole, there has been no marked change in the general method of approach. Essentially this is left to the discretion of the individual teacher. This contrasts strikingly with the American Institute of Biological Sciences Curriculum Study courses, of which there are three versions, each each with its own method of approach (*Nature*, **197**, 4; 1963). The Blue Version, *Molecules to Man*, adopts a biochemical approach which, to some, seems to face the whole process in the wrong direction. The Yellow Version course is developed from the level of cellular organization, and in many ways represents an approach rather in line with some of the English teaching syllabuses. The Green Version adopts an ecological approach, but one very different from the natural history that some of the more successful schools in Britain find attractive.

(2) A broader study of organisms treated as 'going concerns' in relation to each other and their physical environment is encouraged. The type study derived from T. H. Huxley, but lacking much of the depth and rigour with which he originally endowed it, which has dominated advanced secondary school courses in the past, is being largely discarded, and replaced by a more functional treatment. Underlying biological principles are thus receiving more attention than botanical and zoological detail, and biology as a synthesis is replacing botany and zoology as separate disciplines.

(3) Experimental physiology and the elements of biochemistry are receiving a recognition long overdue.

(4) Some attempt is being made, but so far perhaps not sufficiently explicit, to reach a modern position in the field of genetics by drawing together three important lines of evidence, namely, experimental breeding, cytological analysis and biochemical process, and by relating this knowledge to the problems and processes of evolution.

(5) Field work is being more actively stimulated, and there is a distinct trend away from descriptive studies concerned with species lists of well-defined habitats towards more quantitative and analytical studies on a restricted scale. The intention of this approach is to direct attention to the important ecological methods and principles, especially those with economic applications.

(6) Individual or group projects are being included or encouraged and a premium placed on originality.

All these changes appear to be in the right direction, and one welcome general feature is that the syllabuses are now usually accompanied by explanatory notes and suggestions for practical work. It is significant, however, that little assistance has been given, so far, as to how to introduce ATP and DNA, for example, in any but an essentially dogmatic way. There are several problems that must be solved before much of the 'modern biology' can be taught satisfactorily at the secondary school level.

A simple example indicates the general direction of progress. No one would question the desirability of dissection, but some further consideration of its place might make clearer its purpose and the methods to be used in carrying it out. Much time has been wasted in the past in laboriously following dissection guides with the view of uncovering and confirming minute details of structure of little apparent functional consequence. The value of drawing with meticulous care the various stages in the dissection, often without explanatory notes and comments, may also be questioned. Some practice in these techniques is no doubt essential, but dissections less closely directed, covering a wider range of organisms in far less detail and with greater importance attached to the inferences that can be drawn from them, at each stage would provide a better training and prove much more stimulating to the student.

A PALAEOMAGNETIC OBSERVATION OF MIOCENE GEOMAGNETIC SECULAR VARIATION IN OREGON

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THE Earth's magnetic field is subject to fluctuations of several types[1], ranging from variations of external origin, which can last the order of milliseconds, to changes with periods much in excess of the duration of direct observations of the magnetic field, the earliest of which began in London in the sixteenth century[2]. These data from London show a regular clockwise or westward movement of the geomagnetic field around the axial dipole component. Bullard *et al.*[3] and Yukutake[4] have shown that the present westward drift of the geomagnetic field is due to westward drift of non-dipole components of the field at a rate of about 0·2° of longitude per annum. Movement of the main dipole component has not been observed directly[3]. The relatively rapidly moving non-dipole components appear to originate at the core-mantle boundary of the Earth[3].

The regular nature of the London observations was the basis of an assumption which led Watanabe[5] to extrapolate the observed variation of the inclination of the geomagnetic field back several hundred years, for use in dating some baked archaeological remains. Aitken *et al.*[6] have used well-dated pottery kilns and other fired objects to effectively extend observation of the geomagnetic field back to the first century A.D. in Britain. In doing this, Aitken *et al.*[6] showed that extrapolation of observatory data is invalid, and in fact a period of anticlockwise or eastward drifting of the geomagnetic field occurred between A.D. 900 and 1350 in Britain.

Variations of the order already discussed here are generally referred to as secular variations[1]. The character of the secular variation, which is obviously time-dependent, is diversified by geographical variation, as pointed out by Cox[7], who has directed attention to the almost complete lack of geomagnetic secular variation in the Pacific basin at present. A condition of subdued non-dipole components appears to have persisted for at least the past million years, according to palaeomagnetic data from the Pacific region[1]. Despite the complicated variations of the secular change, however, palaeomagnetic results from Plio-Pleistocene rocks indicate that the Earth's magnetic field has been predominantly that of a dipole aligned along the present spin axis for at least the past million years or so[8]. This result, together with the cyclic appearance of the direct observations, and the requirements of the dynamo theory for the origin of the geomagnetic field[9], forms the basis of one of the fundamental assumptions in palaeomagnetism: that is, that the non-axial dipole components of the geomagnetic field

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will average out the spin axis if observation spans a period of about ten thousand years[10]. It has been suggested that larger periods may be needed to accomplish this condition during the Cretaceous period[11].

Obviously the magnitude of the geomagnetic secular variation in remote periods is of basic importance in palaeomagnetism, and in the theory of the origin of the geomagnetic field. Wesink[12] has studied the variation of natural remanent magnetism across seventeen successive Plio-Pleistocene lavas in Iceland. Both normally and reversely magnetized lavas occur in the section, which is likely to span about two million years or more. His observations show that the virtual geomagnetic poles[8], which are the surface expressions of the hypothetical geocentric dipoles creating the observed direction of natural remanent magnetism, have wandered up to 30° from the present spin axis. This is very similar to present conditions where a departure of up to 29° between the observed field and the axial dipole field is disclosed from available observatory records[8].

As a supplement to a regional survey of the palaeomagnetism of the Miocene lavas of southern Oregon[13,14], a section of flat-lying reversely magnetized lavas on Abert Rim (120° 10′ W., 42° 34′ N.) has been sampled in order to examine the secular variation of the geomagnetic field contemporaneous with cooling of the lavas. The precision required necessitated taking about six samples from each lava[15]. A total of 98 cores were taken from 16 successive lavas in a section of 27 lavas totalling about 800 ft. in thickness. Eleven lavas in the section were not sampled, since a minimum separation of 10 ft. between samples could not be maintained because of inaccessibility. A total of 12 cores were taken from four successive baked interbasaltic sediments, averaging about 3 ft. in thickness, near the base of the section. All cores were orientated, while still attached to outcrop, in geographic co-ordinates to an accuracy of 2·5°. The stratigraphic relationships of sampled and unsampled lavas, and interbasaltic zones, are shown as the ordinate in Fig. 1.

Abert Rim is the summit of a typical Basin and Range normal fault, which is little eroded, and steep, forming the eastern side of a large graben. The lavas are mostly Steens Basalt, which is a highly porphyritic dikytaxitic basalt first described by Fuller[16] on Steens Mountain, east of Abert Rim. Sedimentary beds with an Upper Miocene vertebrate fauna[17] lap with a very slight unconformity on to similar basalts exposed on Poker Jim Ridge (120° 10′ W.), a very large fault scarp immediately east of, and parallel to, Abert Rim. Evernden *et al.*[18] have dated tuffs, which interdigitate with the upper part of the Steens basalt, as 15·0 million years old. Watkins[13] has used several factors to show that sections of lavas in southern Oregon accumulated much more rapidly than sections of similar thickness in the Columbia River area bordering Washington and northern Oregon, and therefore sections such as Abert Rim are more suitable for examination of the geomagnetic secular variation than sections to the north, since a minimum time break between lavas is obviously desirable. The time between

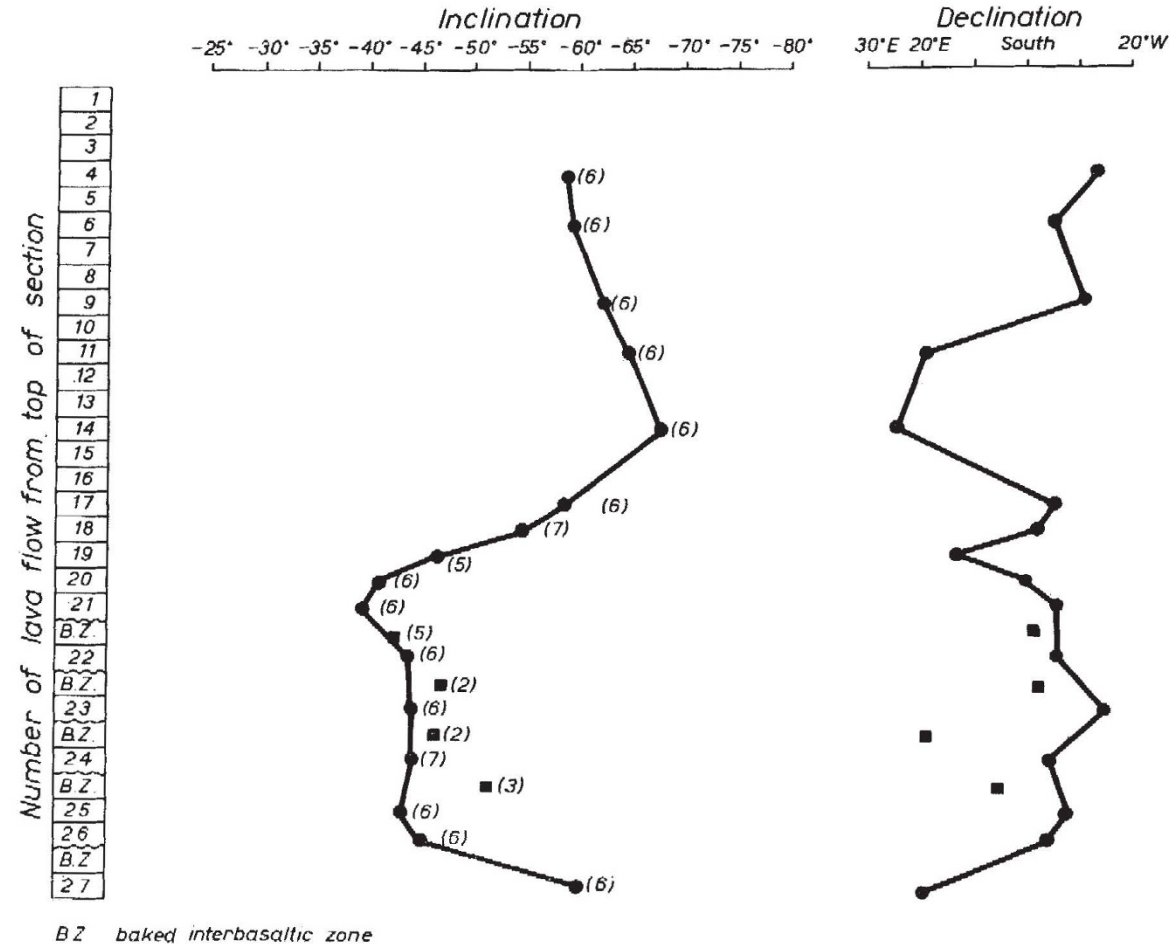


Fig. 1. Mean declination and inclination of natural remanent magnetism of the lavas of Abert Rim, Oregon. Declinations are in geographical azimuths. Inclination is with respect to the horizontal, minus sign indicating an upward direction. Number in brackets refers to the number of specimens taken from each unit. Solid circles refer to natural remanent magnetism from a lava. Solid squares refer to natural remanent magnetism from a baked interbasaltic zone

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successive extrusions of the lavas sampled may average as little as 500 years[19]; but it is not impossible that much longer periods may have occurred between extrusions.

The natural remanent magnetism and the unstable components of natural remanent magnetism of all cores have been examined using a spinner magnetometer[20], and an alternating magnetic field demagnetizing apparatus[21]. Fifteen specimens were subjected to progressive demagnetization in alternating magnetic fields ranging from 25 to 800 oersteds. Very high magnetic stability was evident, and an alternating magnetic field of 50 oersteds, which was considered adequate to remove the small amounts of unstable natural remanent magnetism present, was applied to specimens from all cores. One exception was encountered : a specimen with an anomalously high intensity of magnetization and flat-lying direction of natural remanent magnetism, most likely caused by lightning[22], required application of an alternating magnetic field of 800 oersteds to render agreement of natural remanent magnetism direction with natural remanent magnetism in other specimens from the lava. The progressive demagnetization data are typically those expected in rocks possessing thermal remanent magnetism, although Kobayashi[23] has shown that thermal remanent magnetism is indistinguishable from chemical remanent magnetism on the basis of the comparison of demagnetization data. The natural remanent magnetism of the baked interbasaltic sediments was not subjected to demagnetization experiments, since these specimens are being used in other investigations. The mean direction of natural remanent magnetism and some other statistical parameters[24] are presented in Table 1. Mean declinations and inclinations of natural remanent magnetism are shown in Fig. 1, and the virtual geomagnetic pole for each lava is shown in Fig. 2.

Table 1. Statistical Analysis of Natural Remanent Magnetism of the Lavas of Abert Rim, Oregon

Lava No.	N	$D°$	$I°$	R	k	J
4	6	193·0	−58·8	5·9783	230·4	14·93
6	6	185·0	−59·4	5·9833	299·4	7·59
9	6	190·5	−62·2	5·9858	352·1	3·28
11	7	160·1	−64·6	6·8885	53·8	4·20
14	6	154·3	−67·5	5·9599	124·7	4·45
17	6	184·2	−58·2	5·9683	157·7	5·49
18	7	180·1	−54·1	6·9750	240·0	2·89
19	5	165·1	−46·4	4·7592	16·6	3·26
20	6	178·4	−40·8	5·8941	47·2	4·54
21	6	183·7	−39·4	5·8975	48·2	5·19
B.Z. (*a*)	5	179·5	−42·2	4·9830	—	0·70
22	6	183·0	−43·6	5·9934	757·6	2·53
B.Z. (*b*)	2	180·1	−46·5	1·9824	—	6·01
23	6	192·4	−43·8	5·9439	89·1	2·16
B.Z. (*c*)	2	159·4	−45·8	1·9985	—	4·97
24	7	181·9	−43·8	6·9376	96·1	2·12
B.Z. (*d*)	3	172·3	−50·9	2·6842	—	0·36
25	6	184·8	−42·6	5·9699	142·4	1·91
26	6	181·4	−44·5	5·9791	239·2	1·77
27	6	159·0	−59·0	5·9757	205·8	4·47

Lava No. refers to the number of the lava from the top of the section; N, number of cores taken.

$D°$, Azimuth of mean declination of natural remanent magnetism, with respect to geographical north (angle increases eastwards from north).

$I°$, Mean inclination of natural remanent magnetism with respect to horizontal (— indicates vector above horizontal); R, resultant vector assigning unit vector to each core; k, precision parameter[24] = $(N-1)/(N-R)$; J, mean intensity of magnetization in E.M.U. $cm^{-3} \times 10^{-3}$; *B.Z.*, baked interbasaltic zone.

Polished sections have been made of specimens from all cores, and the titanium-iron oxide minerals have been examined under oil, using a Reichert Zeto-Pol Pan reflexion microscope at a magnification of ×1200. Specimens from all lavas show oxidized forms of titanomagnetite, and ilmenite lamellae in titanomagnetite are

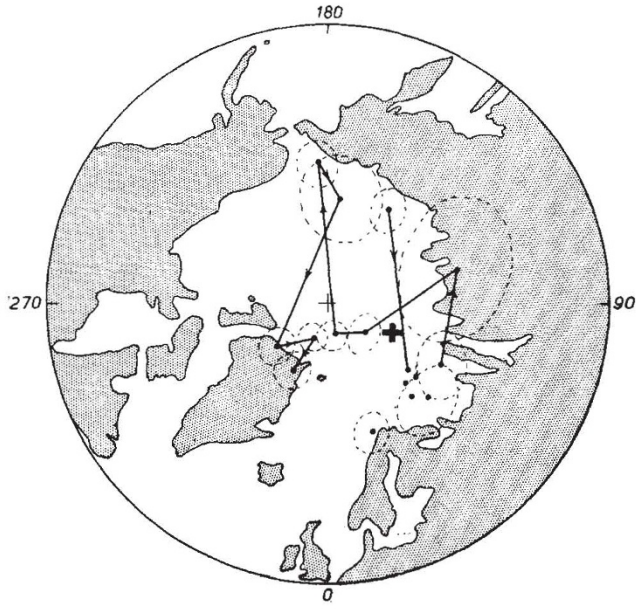


Fig. 2. Virtual geomagnetic poles for the lavas of Abert Rim, Oregon. Virtual geomagnetic poles for successive lavas are joined by a straight line, with the arrow pointing to the younger lava. Ovals of confidence for the 63 per cent confidence cone are included. Some confidence ovals and lines joining successive virtual geomagnetic poles are omitted for clarity. Thick cross, mean virtual geomagnetic pole for entire section, assigning unit vector to each lava; thin cross, north geographical pole. The virtual geomagnetic poles, resulting from south-seeking natural remanent magnetism, have been transposed to the northern hemisphere

common. The state of oxidation of these magnetic minerals is highest near the top of the section, where increased oxidation has led to the common occurrence of pseudobrookite, rutile, and some rutile in a titanium-poor titanomagnetite host, in the youngest two lavas sampled. Further qualitative and quantitative analyses of these minerals are in progress, but the data available so far support correlations observed independently in studies of an Icelandic olivine basalt; the oxidized iron-titanium oxide minerals are magnetically highly stable, and the basalts with the most highly oxidized forms of magnetic constituents possess a higher intensity of magnetization than those with less oxidized minerals[25].

All the available evidence suggests that the geomagnetic field during cooling of the lavas was reversed compared to the present geomagnetic field polarity. This is not surprising since Cox *et al.*[26] have provided strong evidence that the Earth's magnetic field has reversed its polarity several times during the past four million years. The polarity of the sampled baked interbasaltic zones is in all four cases the same as that in the overlying lava, which is magnetically stable. Thus the rarely observed self-reversing mechanism[27] is unlikely to be the cause of the reversed polarity, since the baked sediments and lavas, being so petrologically dissimilar, are unlikely to possess the same self-reversing mechanism, which is likely to be a sensitive function of chemical composition. Since Verhoogen[28] has theorized that oxidation may produce irreproducible self-reversals of polarity in basalts, and Havard[29] has produced near reversals of polarity in titanomagnetites by partial oxidation, as the magnetic minerals in the Abert Rim basalts and the baked interbasaltic zones show signs of oxidation, there remains the apparently remote possibility of self-reversals having occurred to cause the observed reversed polarities. Some evidence for a reversed geomagnetic field being the cause of the polarity in the basalts is forthcoming from palaeomagnetic results from the Ries area, southern Germany: Angenheister and Pohl[30] have reported reversely magnetized volcanic rocks with an age of $14{\cdot}8 \pm 0{\cdot}7$ million years. This is about the age[16,31] of some of the upper

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part of the Steens Basalt type section, where reversely magnetized lavas underlie some normally magnetized lavas, and so it is not impossible that the Ries volcanic material and the Abert Rim lavas cooled during the same reversed geomagnetic polarity epoch.

It can be seen in Figs. 1 and 2 that the five lavas directly associated with the interbasaltic zones have natural remanent magnetism directions which are not significantly different from each other. This situation suggests that since the ambient geomagnetic field defined in younger lavas was apparently subject to some movement, most likely both the interbasaltic sediments and the adjacent lavas may have accumulated during a relatively short period. Thus Geikie's[32] suggestion that two lava flows could be extruded in a very short time and still be separated by a sedimentary interbasaltic zone, resulting possibly from ash falls during violent inter-eruptive activity, may apply to the lavas concerned. The smooth variation of the inclination of natural remanent magnetism shown in Fig. 1 suggests that the top 10 sampled lavas may have cooled during a single cycle of inclination variation, particularly since the first break in the smoothness of the curve occurs at the first sign of a hiatus in the section, in the form of an interbasaltic sediment (Fig. 1). It is strongly emphasized, however, that Figs. 1 and 2 merely show the successive directions of natural remanent magnetism and associated virtual geomagnetic poles in time, and the data do not represent a continual record of the directional behaviour of the ancient magnetic field. The time breaks between the lavas are unknown and are obviously too large to yield a highly detailed record of the secular variation of the ambient geomagnetic field during this part of the Miocene period. As can be readily seen in Fig. 2 particularly, and as must be expected, no meaningful regular trend can be discerned. The results do at least show that the total geomagnetic inclination variation was apparently at least 28°. This is similar to the palaeomagnetic results from Plio-Pleistocene lavas in Iceland, and is of the order of variation indicated from the archaeomagnetic results from Britain[6], which show a total of 22° variation. There appears, then, to be no large difference between the geomagnetic secular variation observed during Historic and Plio-Pleistocene times and that secular variation indicated by reversely magnetized lavas of Miocene age.

Although it is impossible to attribute the data shown in Fig. 1 to any particular component of the Earth's magnetic field[1], a comparison can be made with mathematical models of, or relevant statistical expressions of, the effects of the present geomagnetic field, in order to estimate the similarity of the palaeomagnetic data to that which would be expected from a geomagnetic field of predominantly dipolar character. Irving and Ward[33] have compiled a hypothetical statistical model of the geomagnetic field in the form of a geocentric dipole, perturbed by randomly directed components of constant magnitude which can be considered to be equivalent to non-dipole sources. The latitude dependence of the circular standard deviation (the radius of the circle the centre of which is the true mean direction, and which contains 63 per cent of the individual directions) for the model is very close to the latitude dependence of the circular standard deviation calculated from the 1945 geomagnetic field data[34]. Statistical analysis of the natural remanent magnetism for the Abert Rim lavas, assigning unit to each lava, results in a palaeolatitude of $37° \pm 3{\cdot}5°$ and circular standard deviation of 14·2°. If account is taken of possible tectonic rotation of the section about a vertical axis associated with movement of the orocline[18,35] which is flanked by southern Oregon, the palaeolatitude is correspondingly decreased to $31° \pm 3{\cdot}5°$; and with the circular standard deviation remaining unchanged, a situation which agrees with similar data for the 1945 geomagnetic field, where a latitude of 31° corresponds to a circular standard deviation of 14·2, is arrived at.

This agreement between statistical parameter for the Abert Rim lavas and the 1945 geomagnetic field data does not by any means necessarily prove that the part of the Miocene geomagnetic field concerned was mainly dipolar, since other conditions could be envisaged which would result in statistical similarity; but the agreement is nevertheless consistent with a dipolar geomagnetic field predominating during at least part of the Miocene period.

Finally, it is considered appropriate to mention the potential of the natural remanent magnetism of Abert Rim as a stratigraphic tool. Although in common with many of the thick sections of lava in southern Oregon, only a single polarity is evident[18], and thus detailed surface mapping utilizing polarity differences of lavas such as have been carried out for Icelandic lavas[36] cannot be made, it is obvious from the natural remanent magnetism inclination variation, in particular, that sufficient character exists to enable specific horizons in sections of single magnetic polarity lavas to be traced laterally, if the need arises.

In summary, the investigation of the natural remanent magnetism of the Abert Rim lavas has shown that:

(1) During a period of probable reversed geomagnetic polarity the magnitude of a Miocene secular variation of the inclination of the geomagnetic field was similar to variations in historic and Plio-Pleistocene times.

(2) Analysis of the natural remanent magnetism from the Abert Rim lavas is consistent with a geomagnetic field during part of the Miocene period which is statistically similar to the present geomagnetic field.

(3) The occurrence of interbasaltic horizons may not necessarily represent long periods between the eruption of lavas.

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